

Question ID 310c87fe

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 310c87fe

A cube has a surface area of 54 square meters. What is the volume, in cubic meters, of the cube?

- A. 18
- B. 27
- C. 36
- D. 81

ID: 310c87fe Answer

Correct Answer:

B

Rationale

Choice B is correct. The surface area of a cube with side length s is equal to $6s^2$. Since the surface area is given as 54 square meters, the equation $54 = 6s^2$ can be used to solve for s . Dividing both sides of the equation by 6 yields $9 = s^2$. Taking the square root of both sides of this equation yields $3 = s$ and $-3 = s$. Since the side length of a cube must be a positive value, $s = -3$ can be discarded as a possible solution, leaving $s = 3$. The volume of a cube with side length s is equal to s^3 . Therefore, the volume of this cube, in cubic meters, is 3^3 , or 27.

Choices A, C, and D are incorrect and may result from calculation errors.

Question Difficulty:

Hard

Question ID 459dd6c5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 459dd6c5

Triangles ABC and DEF are similar. Each side length of triangle ABC is 4 times the corresponding side length of triangle DEF . The area of triangle ABC is 270 square inches. What is the area, in square inches, of triangle DEF ?

ID: 459dd6c5 Answer

Correct Answer:
135/8, 16.87, 16.88

Rationale

The correct answer is $\frac{135}{8}$. It's given that triangles ABC and DEF are similar and each side length of triangle ABC is 4 times the corresponding side length of triangle DEF . For two similar triangles, if each side length of the first triangle is k times the corresponding side length of the second triangle, then the area of the first triangle is k^2 times the area of the second triangle. Therefore, the area of triangle ABC is 4^2 , or 16 , times the area of triangle DEF . It's given that the area of triangle ABC is 270 square inches. Let a represent the area, in square inches, of triangle DEF . It follows that 270 is 16 times a , or $270 = 16a$. Dividing both sides of this equation by 16 yields $\frac{270}{16} = a$, which is equivalent to $\frac{135}{8} = a$. Thus, the area, in square inches, of triangle DEF is $\frac{135}{8}$. Note that 135/8, 16.87, and 16.88 are examples of ways to enter a correct answer.

Question Difficulty:
Hard

Question ID 306264ab

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 306264ab

A right triangle has sides of length $2\sqrt{2}$, $6\sqrt{2}$, and $\sqrt{80}$ units. What is the area of the triangle, in square units?

- A. $8\sqrt{2} + \sqrt{80}$
- B. 12
- C. $24\sqrt{80}$
- D. 24

ID: 306264ab Answer

Correct Answer:

B

Rationale

Choice B is correct. The area, A , of a triangle can be found using the formula $A = \frac{1}{2}bh$, where b is the length of the base of the triangle and h is the height of the triangle. It's given that the triangle is a right triangle. Therefore, its base and height can be represented by the two legs. It's also given that the triangle has sides of length $2\sqrt{2}$, $6\sqrt{2}$, and $\sqrt{80}$ units. Since $\sqrt{80}$ units is the greatest of these lengths, it's the length of the hypotenuse. Therefore, the two legs have lengths $2\sqrt{2}$ and $6\sqrt{2}$ units. Substituting these values for b and h in the formula $A = \frac{1}{2}bh$ gives $A = \frac{1}{2}(2\sqrt{2})(6\sqrt{2})$, which is equivalent to $A = 6\sqrt{4}$ square units, or $A = 12$ square units.

Choice A is incorrect. This expression represents the perimeter, rather than the area, of the triangle.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Hard

Question ID c984f1a5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: c984f1a5

A hemisphere is half of a sphere. If a hemisphere has a radius of **27** inches, which of the following is closest to the volume, in cubic inches, of this hemisphere?

- A. 1,500
- B. 6,100
- C. 30,900
- D. 41,200

ID: c984f1a5 Answer

Correct Answer:
D

Rationale

Choice D is correct. The volume, V , of a sphere is given by $V = \frac{4}{3}\pi r^3$, where r is the radius of the sphere. Since a hemisphere is half of a sphere, it follows that the volume, V , of a hemisphere is given by $V = \left(\frac{1}{2}\right)\left(\frac{4}{3}\right)\pi r^3$, or $V = \frac{2}{3}\pi r^3$. Substituting **27** for r in this formula yields $V = \frac{2}{3}\pi(27)^3$, which gives $V = 13,122\pi$, or V is approximately equal to **41,223.98**. Therefore, the choice that is closest to the volume, in cubic inches, of this hemisphere is **41,200**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID f7e626b2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: f7e626b2

The dimensions of a right rectangular prism are 4 inches by 5 inches by 6 inches. What is the surface area, in square inches, of the prism?

- A. 30
- B. 74
- C. 120
- D. 148

ID: f7e626b2 Answer

Rationale

Choice D is correct. The surface area is found by summing the area of each face. A right rectangular prism consists of three pairs of congruent rectangles, so the surface area is found by multiplying the areas of three adjacent rectangles by 2 and adding these products. For this prism, the surface area is equal to $2(4 \cdot 5) + 2(5 \cdot 6) + 2(4 \cdot 6)$, or $2(20) + 2(30) + 2(24)$, which is equal to 148.

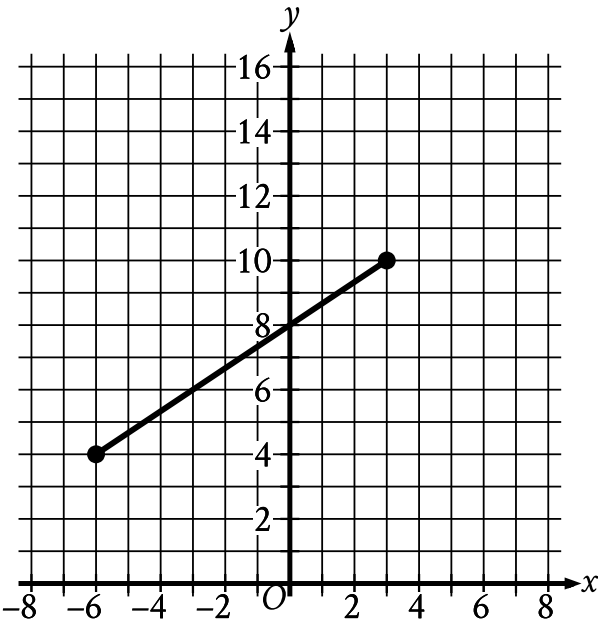
Choice A is incorrect. This is the area of one of the faces of the prism. Choice B is incorrect and may result from adding the areas of three adjacent rectangles without multiplying by 2. Choice C is incorrect. This is the volume, in cubic inches, of the prism.

Question Difficulty:
Hard

Question ID 099526fc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 099526fc



The line segment shown in the xy -plane represents one of the legs of a right triangle. The area of this triangle is $36\sqrt{13}$ square units. What is the length, in units, of the other leg of this triangle?

- A. 12
- B. 24
- C. $3\sqrt{13}$
- D. $18\sqrt{13}$

ID: 099526fc Answer

Correct Answer:
B

Rationale

Choice B is correct. The length of a segment in the xy -plane can be found using the distance formula, $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, where (x_1, y_1) and (x_2, y_2) are the endpoints of the segment. The segment shown has endpoints at $(-6, 4)$ and $(3, 10)$. Substituting $(-6, 4)$ and $(3, 10)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the distance formula yields $\sqrt{(3 - (-6))^2 + (10 - 4)^2}$, or $\sqrt{9^2 + 6^2}$, which is equivalent to $\sqrt{81 + 36}$, or $\sqrt{117}$. Let x represent the length, in units, of the other leg of this triangle. The area, A , of a right triangle can be calculated using the formula $A = \frac{1}{2}bh$, where b and h are the lengths of the legs of the triangle. It's given that the area of the triangle is $36\sqrt{13}$ square units. Substituting $36\sqrt{13}$ for A , $\sqrt{117}$

for b , and x for h in the formula $A = \frac{1}{2}bh$ yields $36\sqrt{13} = \frac{1}{2}(\sqrt{117})(x)$. Multiplying both sides of this equation by 2 yields $72\sqrt{13} = x\sqrt{117}$. Dividing both sides of this equation by $\sqrt{117}$ yields $\frac{72\sqrt{13}}{\sqrt{117}} = x$. Multiplying the numerator and denominator on the left-hand side of this equation by $\sqrt{117}$ yields $\frac{72\sqrt{1,521}}{117} = x$, or $\frac{72(39)}{117} = x$, which is equivalent to $\frac{2,808}{117} = x$, or $24 = x$. Therefore, the length, in units, of the other leg of this triangle is 24 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. $3\sqrt{13}$ is equivalent to $\sqrt{117}$, which is the length, in units, of the line segment shown in the xy -plane, not the length, in units, of the other leg of the triangle.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Hard

Question ID f329442c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: f329442c

Circle A has a radius of $3n$ and circle B has a radius of $129n$, where n is a positive constant. The area of circle B is how many times the area of circle A ?

- A. 43
- B. 86
- C. 129
- D. 1,849

ID: f329442c Answer

Correct Answer:
D

Rationale

Choice D is correct. The area of a circle can be found by using the formula $A = \pi r^2$, where A is the area and r is the radius of the circle. It's given that the radius of circle A is $3n$. Substituting this value for r into the formula $A = \pi r^2$ gives $A = \pi(3n)^2$, or $9\pi n^2$. It's also given that the radius of circle B is $129n$. Substituting this value for r into the formula $A = \pi r^2$ gives $A = \pi(129n)^2$, or $16,641\pi n^2$. Dividing the area of circle B by the area of circle A gives $\frac{16,641\pi n^2}{9\pi n^2}$, which simplifies to 1,849. Therefore, the area of circle B is 1,849 times the area of circle A .

Choice A is incorrect. This is how many times greater the radius of circle B is than the radius of circle A .

Choice B is incorrect and may result from conceptual or calculation errors.

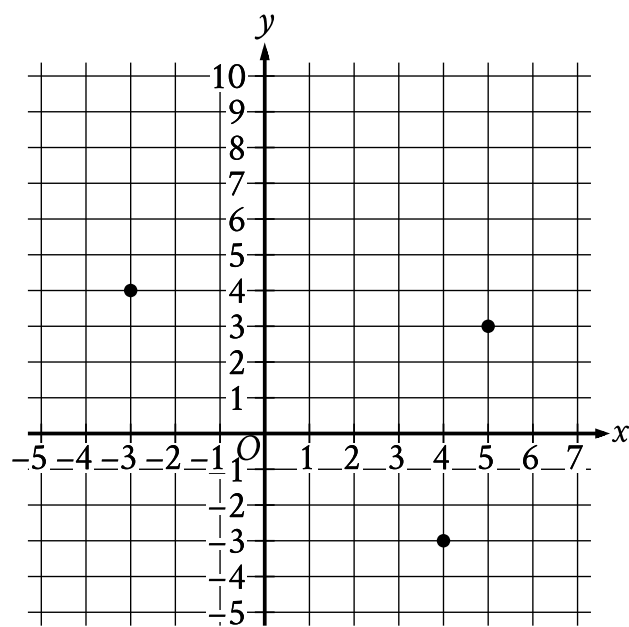
Choice C is incorrect. This is the coefficient on the term that describes the radius of circle B .

Question Difficulty:
Hard

Question ID eb70d2d0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: eb70d2d0



What is the area, in square units, of the triangle formed by connecting the three points shown?

ID: eb70d2d0 Answer

Correct Answer:

24.5, $\frac{49}{2}$

Rationale

The correct answer is **24.5**. It's given that a triangle is formed by connecting the three points shown, which are $(-3, 4)$, $(5, 3)$, and $(4, -3)$. Let this triangle be triangle A. The area of triangle A can be found by calculating the area of the rectangle that circumscribes it and subtracting the areas of the three triangles that are inside the rectangle but outside triangle A. The rectangle formed by the points $(-3, 4)$, $(5, 4)$, $(5, -3)$, and $(-3, -3)$ circumscribes triangle A. The width, in units, of this rectangle can be found by calculating the distance between the points $(5, 4)$ and $(5, -3)$. This distance is $4 - (-3)$, or **7**. The length, in units, of this rectangle can be found by calculating the distance between the points $(5, 4)$ and $(-3, 4)$. This distance is $5 - (-3)$, or **8**. It follows that the area, in square units, of the rectangle is $(7)(8)$, or **56**. One of the triangles that lies inside the rectangle but outside triangle A is formed by the points $(-3, 4)$, $(5, 4)$, and $(5, 3)$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $(5, 4)$ and $(5, 3)$. This distance is $4 - 3$, or **1**. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $(5, 4)$ and $(-3, 4)$. This distance is $5 - (-3)$, or **8**. It follows that the area, in square units, of this triangle is $\frac{1}{2}(8)(1)$, or **4**. A second triangle that lies inside the rectangle but outside triangle A is formed by the points $(4, -3)$, $(5, 3)$, and $(5, -3)$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $(5, 3)$ and $(5, -3)$. This distance is $3 - (-3)$, or **6**. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $(5, -3)$ and $(4, -3)$. This distance is $5 - 4$, or **1**. It follows that the area, in square units, of this triangle is $\frac{1}{2}(1)(6)$, or **3**. The third triangle that lies inside the rectangle but

outside triangle A is formed by the points $(-3, 4)$, $(-3, -3)$, and $(4, -3)$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $(4, -3)$ and $(-3, -3)$. This distance is $4 - (-3)$, or **7**. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $(-3, 4)$ and $(-3, -3)$. This distance is $4 - (-3)$, or **7**. It follows that the area, in square units, of this triangle is $\frac{1}{2}(7)(7)$, or **24.5**. Thus, the area, in square units, of the triangle formed by connecting the three points shown is **$56 - 4 - 3 = 24.5$** , or **24.5**. Note that 24.5 and $49/2$ are examples of ways to enter a correct answer.

Question Difficulty:

Hard

Question ID 93de3f84

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 93de3f84

The volume of right circular cylinder A is 22 cubic centimeters. What is the volume, in cubic centimeters, of a right circular cylinder with twice the radius and half the height of cylinder A?

- A. 11
- B. 22
- C. 44
- D. 66

ID: 93de3f84 Answer

Correct Answer:
C

Rationale

Choice C is correct. The volume of right circular cylinder A is given by the expression $\pi r^2 h$, where r is the radius of its circular base and h is its height. The volume of a cylinder with twice the radius and half the height of cylinder A is given by $\pi (2r)^2 \left(\frac{1}{2}h\right)$, which is equivalent to $4\pi r^2 \left(\frac{1}{2}h\right) = 2\pi r^2 h$. Therefore, the volume is twice the volume of cylinder A, or $2 \times 22 = 44$.

Choice A is incorrect and likely results from not multiplying the radius of cylinder A by 2. Choice B is incorrect and likely results from not squaring the 2 in 2r when applying the volume formula. Choice D is incorrect and likely results from a conceptual error.

Question Difficulty:
Hard

Question ID f243c383

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: f243c383

Two identical rectangular prisms each have a height of **90 centimeters (cm)**. The base of each prism is a square, and the surface area of each prism is $K \text{ cm}^2$. If the prisms are glued together along a square base, the resulting prism has a surface area of $\frac{92}{47}K \text{ cm}^2$. What is the side length, in **cm**, of each square base?

- A. 4
- B. 8
- C. 9
- D. 16

ID: f243c383 Answer

Correct Answer:

B

Rationale

Choice B is correct. Let x represent the side length, in **cm**, of each square base. If the two prisms are glued together along a square base, the resulting prism has a surface area equal to twice the surface area of one of the prisms, minus the area of the two square bases that are being glued together, which yields $2K - 2x^2 \text{ cm}^2$. It's given that this resulting surface area is equal to $\frac{92}{47}K \text{ cm}^2$, so $2K - 2x^2 = \frac{92}{47}K$. Subtracting $\frac{92}{47}K$ from both sides of this equation yields $2K - \frac{92}{47}K - 2x^2 = 0$. This equation can be rewritten by multiplying $2K$ on the left-hand side by $\frac{47}{47}$, which yields $\frac{94}{47}K - \frac{92}{47}K - 2x^2 = 0$, or $\frac{2}{47}K - 2x^2 = 0$. Adding $2x^2$ to both sides of this equation yields $\frac{2}{47}K = 2x^2$. Multiplying both sides of this equation by $\frac{47}{2}$ yields $K = 47x^2$. The surface area K , in cm^2 , of each rectangular prism is equivalent to the sum of the areas of the two square bases and the areas of the four lateral faces. Since the height of each rectangular prism is **90 cm** and the side length of each square base is $x \text{ cm}$, it follows that the area of each square base is $x^2 \text{ cm}^2$ and the area of each lateral face is $90x \text{ cm}^2$. Therefore, the surface area of each rectangular prism can be represented by the expression $2x^2 + 4(90x)$, or $2x^2 + 360x$. Substituting this expression for K in the equation $K = 47x^2$ yields $2x^2 + 360x = 47x^2$. Subtracting $2x^2$ and $360x$ from both sides of this equation yields $0 = 45x^2 - 360x$. Factoring x from the right-hand side of this equation yields $0 = x(45x - 360)$. Applying the zero product property, it follows that $x = 0$ and $45x - 360 = 0$. Adding 360 to both sides of the equation $45x - 360 = 0$ yields $45x = 360$. Dividing both sides of this equation by 45 yields $x = 8$. Since a side length of a rectangular prism can't be **0**, the length of each square base is **8 cm**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Hard

Question ID 1be8b6b2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 1be8b6b2

A right circular cone has a volume of $71,148\pi$ cubic centimeters and the area of its base is $5,929\pi$ square centimeters. What is the slant height, in centimeters, of this cone?

- A. 12
- B. 36
- C. 77
- D. 85

ID: 1be8b6b2 Answer

Correct Answer:
D

Rationale

Choice D is correct. The volume, V , of a right circular cone is given by the formula $V = \frac{1}{3}\pi r^2 h$, where πr^2 is the area of the circular base of the cone and h is the height. It's given that this right circular cone has a volume of $71,148\pi$ cubic centimeters and the area of its base is $5,929\pi$ square centimeters. Substituting $71,148\pi$ for V and $5,929\pi$ for πr^2 in the formula $V = \frac{1}{3}\pi r^2 h$ yields $71,148\pi = (\frac{1}{3})(5,929\pi)(h)$. Dividing each side of this equation by $5,929\pi$ yields $12 = \frac{h}{3}$. Multiplying each side of this equation by 3 yields $36 = h$. Let s represent the slant height, in centimeters, of this cone. A right triangle is formed by the radius, r , height, h , and slant height, s , of this cone, where r and h are the legs of the triangle and s is the hypotenuse. Using the Pythagorean theorem, the equation $r^2 + h^2 = s^2$ represents this relationship. Because $5,929\pi$ is the area of the base and the area of the base is πr^2 , it follows that $5,929\pi = \pi r^2$. Dividing both sides of this equation by π yields $5,929 = r^2$. Substituting 5,929 for r^2 and 36 for h in the equation $r^2 + h^2 = s^2$ yields $5,929 + 36^2 = s^2$, which is equivalent to $5,929 + 1,296 = s^2$, or $7,225 = s^2$. Taking the positive square root of both sides of this equation yields $85 = s$. Therefore, the slant height of the cone is 85 centimeters.

Choice A is incorrect. This is one-third of the height, in centimeters, not the slant height, in centimeters, of this cone.

Choice B is incorrect. This is the height, in centimeters, not the slant height, in centimeters, of this cone.

Choice C is incorrect. This is the radius, in centimeters, of the base, not the slant height, in centimeters, of this cone.

Question Difficulty:
Hard

Question ID dc71597b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: dc71597b

A right circular cone has a volume of $\frac{1}{3}\pi$ cubic feet and a height of 9 feet. What is the radius, in feet, of the base of the cone?

- A. $\frac{1}{3}$
- B. $\frac{1}{\sqrt{3}}$
- C. $\sqrt{3}$
- D. 3

ID: dc71597b Answer

Correct Answer:

A

Rationale

Choice A is correct. The equation for the volume of a right circular cone is $V = \frac{1}{3}\pi r^2h$. It's given that the volume of the right circular cone is $\frac{1}{3}\pi$ cubic feet and the height is 9 feet. Substituting these values for V and h, respectively, gives

$\frac{1}{3}\pi = \frac{1}{3}\pi r^2(9)$. Dividing both sides of the equation by $\frac{1}{3}\pi$ gives $1 = r^2(9)$. Dividing both sides of the equation by 9 gives

$\frac{1}{9} = r^2$. Taking the square root of both sides results in two possible values for the radius, $\sqrt{\left(\frac{1}{9}\right)}$ or $-\sqrt{\left(\frac{1}{9}\right)}$. Since the radius

can't have a negative value, that leaves $\sqrt{\left(\frac{1}{9}\right)}$ as the only possibility. Applying the quotient property of square roots,

$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$, results in $r = \frac{\sqrt{1}}{\sqrt{9}}$, or $r = \frac{1}{3}$.

Choices B and C are incorrect and may result from incorrectly evaluating $\sqrt{\left(\frac{1}{9}\right)}$. Choice D is incorrect and may result from solving $r^2 = 9$ instead of $r^2 = \frac{1}{9}$.

Question Difficulty:

Hard

Question ID 9f934297

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 9f934297

A right rectangular prism has a length of **28 centimeters (cm)**, a width of **15 cm**, and a height of **16 cm**. What is the surface area, **in cm²**, of the right rectangular prism?

ID: 9f934297 Answer

Correct Answer:
2216

Rationale

The correct answer is **2,216**. The surface area of a prism is the sum of the areas of all its faces. A right rectangular prism consists of six rectangular faces, where opposite faces are congruent. It's given that this prism has a length of **28 cm**, a width of **15 cm**, and a height of **16 cm**. Thus, for this prism, there are two faces with area **(28)(15) cm²**, two faces with area **(28)(16) cm²**, and two faces with area **(15)(16) cm²**. Therefore, the surface area, **in cm²**, of the right rectangular prism is **2(28)(15) + 2(28)(16) + 2(15)(16)**, or **2,216**.

Question Difficulty:
Hard

Question ID 5b2b8866

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 5b2b8866

A rectangular poster has an area of **360** square inches. A copy of the poster is made in which the length and width of the original poster are each increased by **20%**. What is the area of the copy, in square inches?

ID: 5b2b8866 Answer

Correct Answer:
2592/5, 518.4

Rationale

The correct answer is **518.4**. It's given that the area of the original poster is **360** square inches. Let ℓ represent the length, in inches, of the original poster, and let w represent the width, in inches, of the original poster. Since the area of a rectangle is equal to its length times its width, it follows that $360 = \ell w$. It's also given that a copy of the poster is made in which the length and width of the original poster are each increased by **20%**. It follows that the length of the copy is the length of the original poster plus **20%** of the length of the original poster, which is equivalent to $\ell + \frac{20}{100}\ell$ inches. This length can be rewritten as $\ell + 0.2\ell$ inches, or 1.2ℓ inches. Similarly, the width of the copy is the width of the original poster plus **20%** of the width of the original poster, which is equivalent to $w + \frac{20}{100}w$ inches. This width can be rewritten as $w + 0.2w$ inches, or $1.2w$ inches. Since the area of a rectangle is equal to its length times its width, it follows that the area, in square inches, of the copy is equal to $(1.2\ell)(1.2w)$, which can be rewritten as $(1.2)(1.2)(\ell w)$. Since $360 = \ell w$, the area, in square inches, of the copy can be found by substituting **360** for ℓw in the expression $(1.2)(1.2)(\ell w)$, which yields $(1.2)(1.2)(360)$, or **518.4**. Therefore, the area of the copy, in square inches, is **518.4**.

Question Difficulty:
Hard

Question ID 8c1aa743

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 8c1aa743

Rectangles $ABCD$ and $EFGH$ are similar. The length of each side of $EFGH$ is 6 times the length of the corresponding side of $ABCD$. The area of $ABCD$ is 54 square units. What is the area, in square units, of $EFGH$?

- A. 9
- B. 36
- C. 324
- D. $1,944$

ID: 8c1aa743 Answer

Correct Answer:
D

Rationale

Choice D is correct. The area of a rectangle is given by bh , where b is the length of the base of the rectangle and h is its height. Let x represent the length, in units, of the base of rectangle $ABCD$, and let y represent its height, in units. Substituting x for b and y for h in the formula bh yields xy . Therefore, the area, in square units, of $ABCD$ can be represented by the expression xy . It's given that the length of each side of $EFGH$ is 6 times the length of the corresponding side of $ABCD$. Therefore, the length, in units, of the base of $EFGH$ can be represented by the expression $6x$, and its height, in units, can be represented by the expression $6y$. Substituting $6x$ for b and $6y$ for h in the formula bh yields $(6x)(6y)$, which is equivalent to $36xy$. Therefore, the area, in square units, of $EFGH$ can be represented by the expression $36xy$. It's given that the area of $ABCD$ is 54 square units. Since xy represents the area, in square units, of $ABCD$, substituting 54 for xy in the expression $36xy$ yields $36(54)$, or $1,944$. Therefore, the area, in square units, of $EFGH$ is $1,944$.

Choice A is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{\frac{1}{6}}$, not 6 , times the length of the corresponding side of $ABCD$.

Choice B is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{\frac{2}{3}}$, not 6 , times the length of the corresponding side of $ABCD$.

Choice C is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{6}$, not 6 , times the length of the corresponding side of $ABCD$.

Question Difficulty:
Hard

Question ID 6708546e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 6708546e

Parallelogram $ABCD$ is similar to parallelogram $PQRS$. The length of each side of parallelogram $PQRS$ is 2 times the length of its corresponding side of parallelogram $ABCD$. The area of parallelogram $ABCD$ is 5 square centimeters. What is the area, in square centimeters, of parallelogram $PQRS$?

- A. 7
- B. 10
- C. 20
- D. 25

ID: 6708546e Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that parallelogram $ABCD$ is similar to parallelogram $PQRS$. When two parallelograms are similar, if the scale factor between their corresponding side lengths is k , the scale factor between their areas is k^2 . It's given that the length of each side of parallelogram $PQRS$ is 2 times the length of its corresponding side of parallelogram $ABCD$, so the scale factor between their corresponding side lengths is 2 . It follows that the scale factor between their areas is 2^2 , or 4 . It's given that the area, in square centimeters, of parallelogram $ABCD$ is 5 . It follows that the area, in square centimeters, of parallelogram $PQRS$ is $5(4)$, or 20 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID b0dc920d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: b0dc920d

A manufacturer determined that right cylindrical containers with a height that is 4 inches longer than the radius offer the optimal number of containers to be displayed on a shelf. Which of the following expresses the volume, V , in cubic inches, of such containers, where r is the radius, in inches?

- A. $V = 4\pi^3$
- B. $V = \pi(2r)^3$
- C. $V = \pi^2 + 4\pi$
- D. $V = \pi^3 + 4\pi^2$

ID: b0dc920d Answer

Correct Answer:
D

Rationale

Choice D is correct. The volume, V , of a right cylinder is given by the formula $V = \pi^2h$, where r represents the radius of the base of the cylinder and h represents the height. Since the height is 4 inches longer than the radius, the expression $r + 4$ represents the height of each cylindrical container. It follows that the volume of each container is represented by the equation $V = \pi^2(r + 4)$. Distributing the expression π^2 into each term in the parentheses yields $V = \pi^3 + 4\pi^2$.

Choice A is incorrect and may result from representing the height as $4r$ instead of $r + 4$. Choice B is incorrect and may result from representing the height as $2r$ instead of $r + 4$. Choice C is incorrect and may result from representing the volume of a right cylinder as $V = \pi h$ instead of $V = \pi^2h$.

Question Difficulty:
Hard

Question ID 899c6042

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 899c6042

A right circular cone has a height of **22 centimeters (cm)** and a base with a diameter of **6 cm**. The volume of this cone is $n\pi \text{ cm}^3$. What is the value of n ?

ID: 899c6042 Answer

Correct Answer:
66

Rationale

The correct answer is **66**. It's given that the right circular cone has a height of **22 centimeters (cm)** and a base with a diameter of **6 cm**. Since the diameter of the base of the cone is **6 cm**, the radius of the base is **3 cm**. The volume V , **in cm³**, of a right circular cone can be found using the formula $V = \frac{1}{3}\pi r^2 h$, where h is the height, **in cm**, and r is the radius, **in cm**, of the base of the cone. Substituting **22** for h and **3** for r in this formula yields $V = \frac{1}{3}\pi(3)^2(22)$, or $V = 66\pi$. Therefore, the volume of the cone is **66π cm³**. It's given that the volume of the cone is $n\pi \text{ cm}^3$. Therefore, the value of n is **66**.

Question Difficulty:
Hard

Question ID ba8ca563

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: ba8ca563

A cube has a volume of **474,552** cubic units. What is the surface area, in square units, of the cube?

ID: ba8ca563 Answer

Correct Answer:
36504

Rationale

The correct answer is **36,504**. The volume of a cube can be found using the formula $V = s^3$, where s represents the edge length of a cube. It's given that this cube has a volume of **474,552** cubic units. Substituting **474,552** for V in $V = s^3$ yields **474,552** = s^3 . Taking the cube root of both sides of this equation yields **78** = s . Thus, the edge length of the cube is **78** units. Since each face of a cube is a square, it follows that each face has an edge length of **78** units. The area of a square can be found using the formula $A = s^2$. Substituting **78** for s in this formula yields $A = 78^2$, or $A = 6,084$. Therefore, the area of one face of this cube is **6,084** square units. Since a cube has **6** faces, the surface area, in square units, of this cube is **6(6,084)**, or **36,504**.

Question Difficulty:
Hard

Question ID 9fec9d49

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 9fec9d49

The floor of a ballroom has an area of **600** square meters. An architect creates a scale model of the floor of the ballroom, where the length of each side of the model is $\frac{1}{10}$ times the length of the corresponding side of the actual floor of the ballroom. What is the area, in square meters, of the scale model?

- A. **6**
- B. **10**
- C. **60**
- D. **150**

ID: 9fec9d49 Answer

Correct Answer:

A

Rationale

Choice A is correct. It's given that the length of each side of a scale model is $\frac{1}{10}$ times the length of the corresponding side of the actual floor of a ballroom. Therefore, the area of the scale model is $\left(\frac{1}{10}\right)^2$, or $\frac{1}{100}$, times the area of the actual floor of the ballroom. It's given that the area of the floor of the ballroom is **600** square meters. Therefore, the area, in square meters, of the scale model is $\left(\frac{1}{100}\right)(600)$, or **6**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

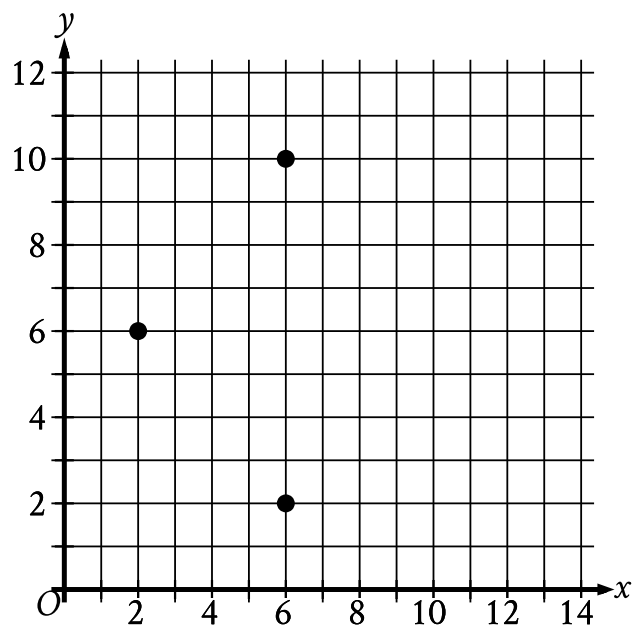
Question Difficulty:

Hard

Question ID b2528e6b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: b2528e6b



The three points shown define a circle. The circumference of this circle is $k\pi$, where k is a constant. What is the value of k ?

ID: b2528e6b Answer

Correct Answer:
8

Rationale

The correct answer is 8. It's given that the three points shown define a circle, so the center of that circle is an equal distance from each of the three points. The point (6, 6) is halfway between the points (6, 2) and (6, 10), and is a distance of 4 units from each of those two points. The point (6, 6) is also a distance of 4 units from (2, 6). Because the point (6, 6) is the same distance from all three points shown, it must be the center of the circle. Since that distance is 4, it follows that the radius of the circle is 4. The circumference of a circle with radius r is equal to $2\pi r$. It follows that the circumference of the given circle is $2\pi(4)$, or 8π . It's given that the circumference of the circle is $k\pi$. Therefore, the value of k is 8.

Question Difficulty:
Hard

Question ID 502d9690

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 502d9690

Rectangle $ABCD$ is similar to rectangle $EFGH$. The area of rectangle $ABCD$ is **648** square inches, and the area of rectangle $EFGH$ is **72** square inches. The length of the longest side of rectangle $ABCD$ is **36** inches. What is the length, in inches, of the longest side of rectangle $EFGH$?

- A. 4
- B. 9
- C. 12
- D. 36

ID: 502d9690 Answer

Correct Answer:

C

Rationale

Choice C is correct. It's given that rectangle $ABCD$ is similar to rectangle $EFGH$. Therefore, if the length of each side of rectangle $ABCD$ is k times the length of the corresponding side of rectangle $EFGH$, then the area of rectangle $ABCD$ is k^2 times the area of rectangle $EFGH$. It's given that the area of rectangle $ABCD$ is **648** square inches and the area of rectangle $EFGH$ is **72** square inches. It follows that $k^2 = \frac{648}{72}$, or $k^2 = 9$. Taking the square root of each side of this equation yields $k = \sqrt{9}$, or $k = 3$. It follows that the length of each side of rectangle $ABCD$ is **3** times the length of the corresponding side of rectangle $EFGH$. It's given that the length of the longest side of rectangle $ABCD$ is **36** inches. Therefore, **36** inches is **3** times the length of the longest side of rectangle $EFGH$, and the longest side of rectangle $EFGH$ is equal to $\frac{36}{3}$, or **12**, inches.

Choice A is incorrect. This is the length, in inches, of the longest side of a rectangle with side lengths that are $\frac{1}{9}$ the corresponding side lengths of rectangle $ABCD$, rather than a rectangle with an area that is $\frac{1}{9}$ the area of rectangle $ABCD$.

Choice B is incorrect. This is the factor by which the area of rectangle $ABCD$ is larger than the area of rectangle $EFGH$, not the length, in inches, of the longest side of rectangle $EFGH$.

Choice D is incorrect. This is the length, in inches, of the longest side of rectangle $ABCD$, not rectangle $EFGH$.

Question Difficulty:

Hard

Question ID 167aff9e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 167aff9e

Right rectangular prism X is similar to right rectangular prism Y. The surface area of right rectangular prism X is **58 square centimeters (cm^2)**, and the surface area of right rectangular prism Y is **1,450 cm^2** . The volume of right rectangular prism Y is **1,250 cubic centimeters (cm^3)**. What is the sum of the volumes, **in cm^3** , of right rectangular prism X and right rectangular prism Y?

ID: 167aff9e Answer

Correct Answer:

1260

Rationale

The correct answer is **1,260**. Since it's given that prisms X and Y are similar, all the linear measurements of prism Y are **k** times the respective linear measurements of prism X, where **k** is a positive constant. Therefore, the surface area of prism Y is **k^2** times the surface area of prism X and the volume of prism Y is **k^3** times the volume of prism X. It's given that the surface area of prism Y is **1,450 cm^2** , and the surface area of prism X is **58 cm^2** , which implies that **$1,450 = 58k^2$** . Dividing both sides of this equation by **58** yields **$\frac{1,450}{58} = k^2$** , or **$k^2 = 25$** . Since **k** is a positive constant, **$k = 5$** . It's given that the volume of prism Y is **1,250 cm^3** . Therefore, the volume of prism X is equal to **$\frac{1,250}{k^3} \text{ cm}^3$** , which is equivalent to **$\frac{1,250}{5^3} \text{ cm}^3$** , or **10 cm^3** . Thus, the sum of the volumes, in **cm^3** , of the two prisms is **$1,250 + 10$** , or **1,260**.

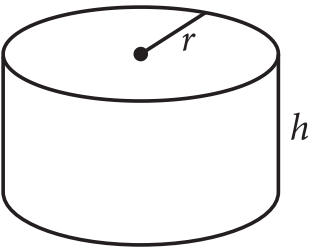
Question Difficulty:

Hard

Question ID a07ed090

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: a07ed090



The figure shown is a right circular cylinder with a radius of r and height of h . A second right circular cylinder (not shown) has a volume that is **392** times as large as the volume of the cylinder shown. Which of the following could represent the radius R , in terms of r , and the height H , in terms of h , of the second cylinder?

- A. $R = 8r$ and $H = 7h$
- B. $R = 8r$ and $H = 49h$
- C. $R = 7r$ and $H = 8h$
- D. $R = 49r$ and $H = 8h$

ID: a07ed090 Answer

Correct Answer:
C

Rationale

Choice C is correct. The volume of a right circular cylinder is equal to $\pi a^2 b$, where a is the radius of a base of the cylinder and b is the height of the cylinder. It's given that the cylinder shown has a radius of r and a height of h . It follows that the volume of the cylinder shown is equal to $\pi r^2 h$. It's given that the second right circular cylinder has a radius of R and a height of H . It follows that the volume of the second cylinder is equal to $\pi R^2 H$. Choice C gives $R = 7r$ and $H = 8h$. Substituting $7r$ for R and $8h$ for H in the expression that represents the volume of the second cylinder yields $\pi(7r)^2(8h)$, or $\pi(49r^2)(8h)$, which is equivalent to $\pi(392r^2h)$, or $392(\pi r^2h)$. This expression is equal to **392** times the volume of the cylinder shown, $\pi r^2 h$. Therefore, $R = 7r$ and $H = 8h$ could represent the radius R , in terms of r , and the height H , in terms of h , of the second cylinder.

Choice A is incorrect. Substituting $8r$ for R and $7h$ for H in the expression that represents the volume of the second cylinder yields $\pi(8r)^2(7h)$, or $\pi(64r^2)(7h)$, which is equivalent to $\pi(448r^2h)$, or $448(\pi r^2h)$. This expression is equal to **448**, not **392**, times the volume of the cylinder shown.

Choice B is incorrect. Substituting $8r$ for R and $49h$ for H in the expression that represents the volume of the second cylinder yields $\pi(8r)^2(49h)$, or $\pi(64r^2)(49h)$, which is equivalent to $\pi(3,136r^2h)$, or $3,136(\pi r^2h)$. This expression is equal to **3,136**, not **392**, times the volume of the cylinder shown.

Choice D is incorrect. Substituting $49r$ for R and $8h$ for H in the expression that represents the volume of the second cylinder yields $\pi(49r)^2(8h)$, or $\pi(2,401r^2)(8h)$, which is equivalent to $\pi(19,208r^2h)$, or $19,208(\pi r^2h)$. This expression is equal to

19,208, not 392, times the volume of the cylinder shown.

Question Difficulty:

Hard

Question ID 9966235e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 9966235e

A cube has an edge length of **68** inches. A solid sphere with a radius of **34** inches is inside the cube, such that the sphere touches the center of each face of the cube. To the nearest cubic inch, what is the volume of the space in the cube not taken up by the sphere?

- A. **149,796**
- B. **164,500**
- C. **190,955**
- D. **310,800**

ID: 9966235e Answer

Correct Answer:
A

Rationale

Choice A is correct. The volume of a cube can be found by using the formula $V = s^3$, where V is the volume and s is the edge length of the cube. Therefore, the volume of the given cube is $V = 68^3$, or **314,432** cubic inches. The volume of a sphere can be found by using the formula $V = \frac{4}{3}\pi r^3$, where V is the volume and r is the radius of the sphere. Therefore, the volume of the given sphere is $V = \frac{4}{3}\pi(34)^3$, or approximately **164,636** cubic inches. The volume of the space in the cube not taken up by the sphere is the difference between the volume of the cube and volume of the sphere. Subtracting the approximate volume of the sphere from the volume of the cube gives $314,432 - 164,636 = 149,796$ cubic inches.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID deff8a2f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: deff8a2f

The circumference of the base of a right circular cylinder is 20π meters, and the height of the cylinder is 6 meters. What is the volume, in cubic meters, of the cylinder?

- A. 60π
- B. 120π
- C. 600π
- D. $2,400\pi$

ID: deff8a2f Answer

Correct Answer:
C

Rationale

Choice C is correct. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. It's given that a right circular cylinder has a height of 6 meters. Therefore, $h = 6$. It's also given that the right circular cylinder has a base with a circumference of 20π meters. The circumference, C , of a circle is given by $C = 2\pi r$, where r is the radius of the circle. Substituting 20π for C in the formula $C = 2\pi r$ yields $20\pi = 2\pi r$. Dividing each side of this equation by 2π yields $10 = r$. Substituting 10 for r and 6 for h in the formula $V = \pi r^2 h$ yields $V = \pi(10)^2(6)$, or $V = 600\pi$. Therefore, the volume, in cubic meters, of the cylinder is 600π .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the lateral surface area, not the volume, of the cylinder.

Choice D is incorrect. This is the result of using the diameter, not the radius, for the value of r in the formula $V = \pi r^2 h$.

Question Difficulty:
Hard

Question ID 983412ea

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 983412ea

A right square prism has a height of **14** units. The volume of the prism is **2,016** cubic units. What is the length, in units, of an edge of the base?

ID: 983412ea Answer

Correct Answer:

12

Rationale

The correct answer is **12**. The volume, V , of a right square prism can be calculated using the formula $V = s^2h$, where s represents the length of an edge of the base and h represents the height of the prism. It's given that the volume of the prism is **2,016** cubic units and the height is **14** units. Substituting **2,016** for V and **14** for h in the formula $V = s^2h$ yields $2,016 = (s^2)(14)$. Dividing both sides of this equation by **14** yields $144 = s^2$. Taking the square root of both sides of this equation yields two solutions: $-12 = s$ and $12 = s$. The length can't be negative, so $12 = s$. Therefore, the length, in units, of an edge of the base is **12**.

Question Difficulty:

Hard

Question ID e5c57163

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: e5c57163

Square A has side lengths that are **166** times the side lengths of square B. The area of square A is **k** times the area of square B. What is the value of **k** ?

ID: e5c57163 Answer

Correct Answer:
27556

Rationale

The correct answer is **27,556**. The area of a square is **s^2** , where **s** is the side length of the square. Let **x** represent the length of each side of square B. Substituting **x** for **s** in **s^2** yields **x^2** . It follows that the area of square B is **x^2** . It's given that square A has side lengths that are **166** times the side lengths of square B. Since **x** represents the length of each side of square B, the length of each side of square A can be represented by the expression **$166x$** . It follows that the area of square A is **$(166x)^2$** , or **$27,556x^2$** . It's given that the area of square A is **k** times the area of square B. Since the area of square A is equal to **$27,556x^2$** , and the area of square B is equal to **x^2** , an equation representing the given statement is **$27,556x^2 = kx^2$** . Since **x** represents the length of each side of square B, the value of **x** must be positive. Therefore, the value of **x^2** is also positive, so it does not equal **0**. Dividing by **x^2** on both sides of the equation **$27,556x^2 = kx^2$** yields **$27,556 = k$** . Therefore, the value of **k** is **27,556**.

Question Difficulty:
Hard